EVOLVING CONVOLUTIONAL FILTERS

with Differential Evolution

Artificial Inteligence — ex Report

Authors: [Lateef Hanus] & [Michal chojnacki]

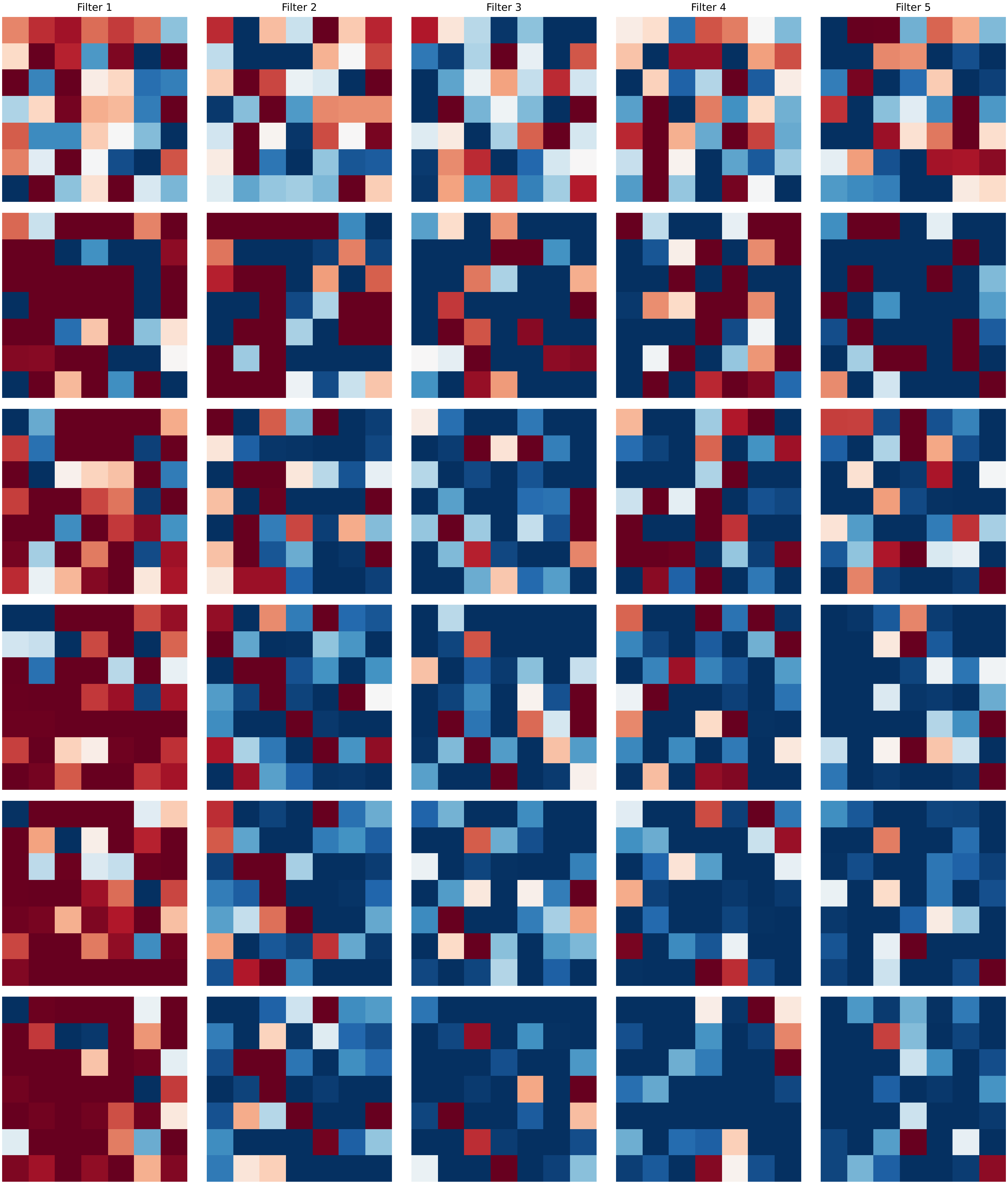
Date: June 15, 2026

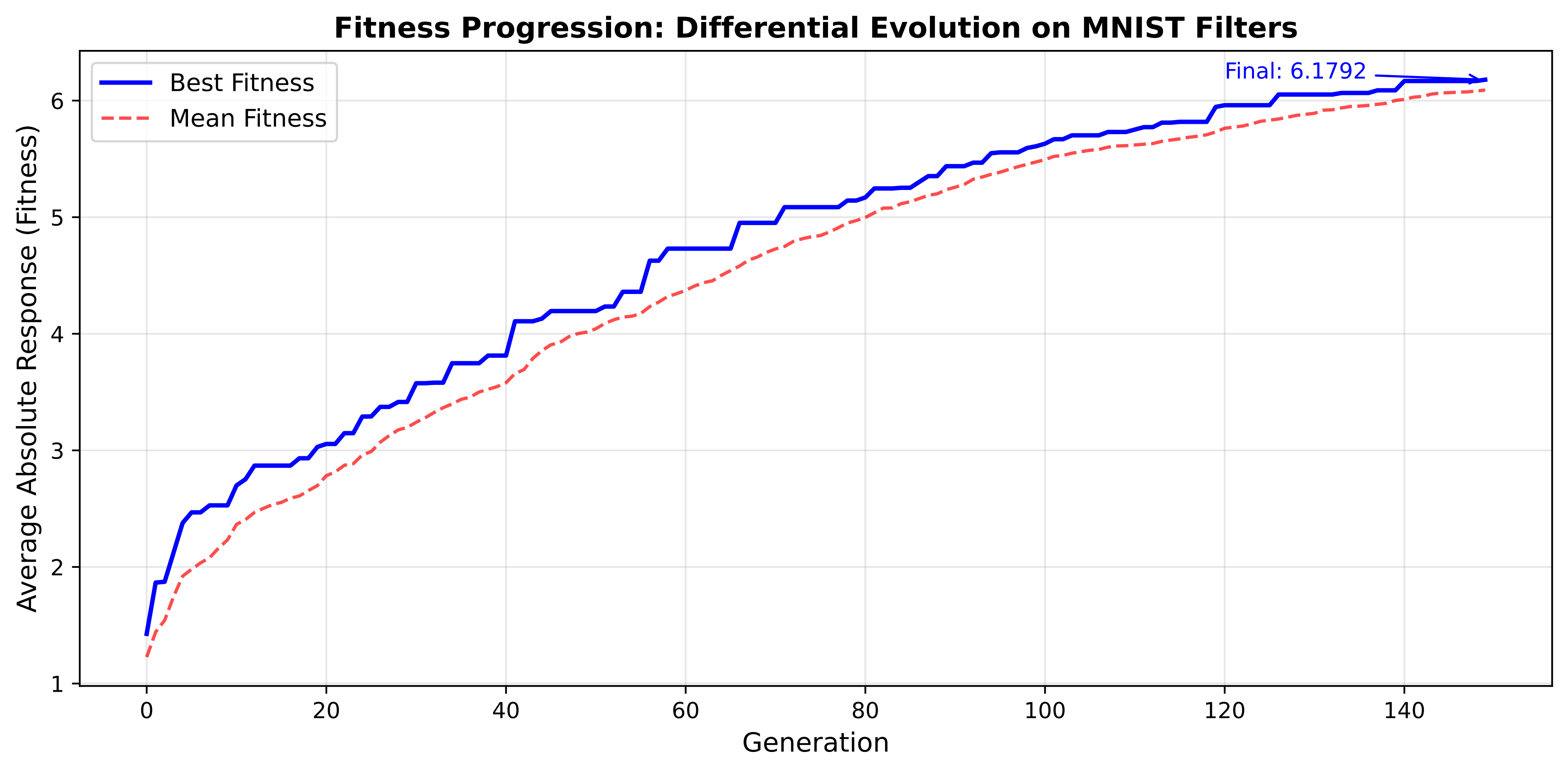
Code Repository:

https://github.com/latef9326/dimensionality_reudction

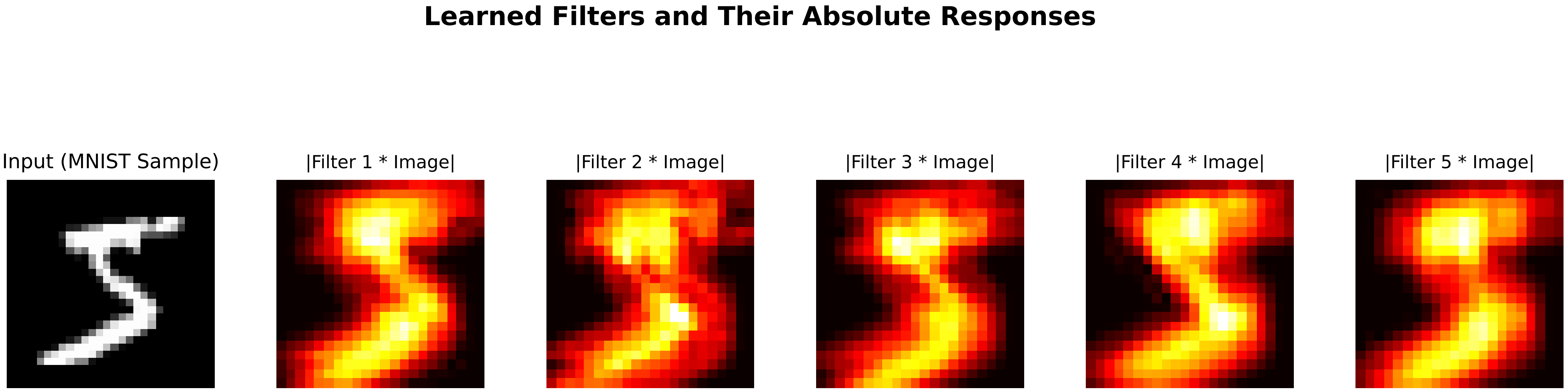
```
ALGORITHM: Differential Evolution (DE/rand/1/bin)

Differential Evolution is a population-based metaheuristic introduced by
Storn & Price (1997). Unlike traditional Genetic Algorithms, DE does NOT use:
  • Crossover operators like one-point or uniform crossover
  • Bit-string representations or binary encoding
  • Fitness-proportionate selection

Instead, DE operates on real-valued vectors and uses:

1. MUTATION (DE/rand/1):
   v_i = x_r1 + F · (x_r2 − x_r3)
   Three distinct random individuals (r1, r2, r3) are selected.
   The difference vector (x_r2 − x_r3) is scaled by F and added to x_r1.

2. CROSSOVER (Binomial):
   Each gene in the trial vector is inherited from either the mutant
   or the target, with probability CR. One gene is forced from the
   mutant to ensure diversity.

3. SELECTION (Greedy):
   The trial vector replaces the target only if it has BETTER fitness.
   This is deterministic, not probabilistic.

HYPERPARAMETERS:
  • Population size (μ):        30
  • Differential weight (F):    0.8
  • Crossover rate (CR):        0.9
  • Gene bounds:                [-1.0, 1.0]
  • Elitism:                    Enabled (best preserved)
  • Generations:                150
  • Genome size:                245 (5 filters × 7×7 weights)
```

Evolution of 5 Convolutional Filters (7×7) Over Generations
Differential Evolution on MNIST





Fitness Progression: Differential Evolution on MNIST Filters
Best Fitness
Mean Fitness
Final: 6.1792
Average Absolute Response (Fitness)
Generation
1
2
3
4
5
6
0
20
40
60
80
100
120
140


Learned Filters and Their Absolute Responses
Input (MNIST Sample)
|Filter 1 * Image|
|Filter 2 * Image|
|Filter 3 * Image|
|Filter 4 * Image|
|Filter 5 * Image|

AUTHOR CONTRIBUTIONS

This assignment was completed collaboratively by a team of two students.

AUTHOR 1: [Lateef Hanus]

- Designed the Differential Evolution algorithm structure
- Implemented mutation (DE/rand/1) and binomial crossover
- Set up the fitness function using PyTorch conv2d operations
- Tuned hyperparameters (F, CR, population size)
- Ran experiments and collected fitness trajectories

AUTHOR 2: [Michal Chojnacki]

- Implemented MNIST data loading and preprocessing pipeline
- Designed and coded all visualization functions
- Created the filter evolution grid and fitness progression plots
- Generated the PDF report with matplotlib.backends.backend_pdf
- Wrote the algorithm description and contribution documentation

Both authors contributed equally to debugging, result analysis, and report writing. All code was reviewed by both team members.

NOTE ON TEAMWORK POLICY:
This submission is from a team of two as required. If working alone due to exceptional circumstances, pre-approval from the instructor must be obtained per the syllabus.